

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 2 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 2

Subject Code & Name: 24AF1PHYBS202 - ENGINEERING PHYSICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 O.W.U. is the unit of
a. Absorption b. Reverberation time coefficient c. Loudness d. Absorption
- 2 How does ionic polarization occur?
a. Splitting of ions b. Passing magnetic field c. Displacement of cations and anions d. Never occurs
- 3 In the Newton's rings experiment, if the radius of curvature of the lens increases, the diameter of the rings will
a. Increase b. decrease c. Remain the same d. Become zero
- 4 If the light is made incident on any transparent medium at the polarizing angle, the reflected light is
a. unpolarised b. Plane polarised c. partially Polarised d. none of these
- 5 What is the life time of electron in metastable state?
a. 10^{-3} sec b. 10^{-5} Sec c. 10^{-8} sec d. 10^{-7} sec
- 6 What is the term for the maximum angle at which light can enter an optical fiber and still be guided through it?
a. Refraction angle b. Critical angle c. Acceptance angle d. Numerical aperture
- 7 The concept of matter wave was suggested by
a. Heisenberg b. De- Broglie c. Schrodinger d. Laplace
- 8 What is the fundamental unit of information in quantum computing?
a. Qubit b. Bit c. Byte d. Quantum Byte
- 9 Which of the following acts as quenching in Geiger Muller counter?
a. Alcohol b. Argon c. Krypton d. Hydrogen
- 10 What is the smallest repeating unit in a crystal lattice that, when repeated in three dimensions, generates the entire crystal structure?
a. Unit Cell b. Space lattice c. Basis d. Miller indices

11. What is the typical size range for nanomaterials (in at least one dimension)?
 a. 1-10 micrometers b. 10- 1000 nanometer c. 1-100 nanometers d. 0.1-1 nanometer
12. The magnetic lines of force cannot penetrate the body of a superconductor, a phenomenon is known as
 a. Isotopic effect b. BCS Theory c. Meissner effect d. London theory
- Q2. Solve the following.
- A) What is Magnetostriction effect? Describe the production of ultrasonic waves by using Magnetostriction method. 6
- B) Explain the types of dielectric polarization. 6
- The volume of an auditorium is 12000 m^3 . Its reverberation time is 1.5 sec. If the average absorption coefficient of interior surface is $0.4 \text{ Sabine m}^{-2}$. Find the area of interior surface.
- Q3. Solve the following.
- A) Explain the construction and working of semiconductor diode laser. 6
- B) Explain the phenomenon of interference in thin films in reflected light. 6
- Q4. Solve Any Two of the following.
- A) State Heisenberg's Uncertainty Principle. Using this principle, if uncertainty in the location of a particle is equal to its De-Broglie wavelength. Show that the uncertainty in its velocity is equal to its velocity. 6
- B) Derive Schrodinger's time dependent wave equation. 6
- C) Write a note on quantum computing. What is difference between bits and qubits? 6
- Q5. Solve Any Two of the following.
- A) Define atomic radius. Derive an expression for atomic radii for SC, BCC and FCC. 6
- B) What are Miller Indices? Explain procedure for find Miller Indices. Find the miller Indices of a set of parallel planes which makes intercepts in the ration $3a:4b$ on X and Y axis and are parallel to Z axes. 6
- C) State different properties of α , β and γ rays. 6
- Q6. Solve Any Two of the following.
- A) Distinguish between Type –I and Type – II superconductors along with examples. 6
- B) Derive an expression for Hall voltage and Hall coefficient with neat labeled diagram. 6
- C) What is nanomaterial? Explain top-down and Bottom-up approach. 6

*** End ***